# THE FACTORIZATION PROPERTY FOR EXT ONE ANNIHILATION IS PRESERVED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- The equivalence of the annihilation condition with the factorization of multiplication maps through the differential, and the two-term representation. invoked at statement lines 22556-22560 and proof lines 22564-22566. Supplies the explicit two-term model on which the base change is performed.
- Base change of a two-term complex and the truncation appearing in the conclusion. invoked at proof lines 22567-22569. Identifies the truncated base change with the base change of the representing complex.
- Linearity of the factoring maps over the base ring. used at proof lines 22572-22574. Extends the factorization from the generators of I to the whole ideal IR'.

*Source reference.* Original lemma 22452: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22452–22452. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 22520: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22520–22520. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

**Lemma 0.2.** *Let $I$ be an ideal of a ring $R$. Let $K$ be an object of $D(R)$ with $H^i(K) = 0$ for $i \notin \{-1, 0\}$. Let $R \to R'$ be a ring map. If $K$ satisfies the equivalent conditions (1), (2), and (3) of Lemma 22452 with respect to $(R, I)$, then*

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*$\tau_{\geq -1}(K \otimes^{\mathbf{L}}_R R')$ satisfies the equivalent conditions (1), (2), and (3) of Lemma 22452 with respect to $(R', IR')$*

*Proof.* We may assume $K$ is represented by a two term complex $K^{-1} \to K^0$ with $K^0$ free such that for any $a \in I$ the map $a : K^{-1} \to K^{-1}$ is equal to $h_a \circ d_K^{-1}$ for some map $h_a : K^0 \to K^{-1}$. By Lemma 22520 we see that $\tau_{\geq -1}(K \otimes^{\mathbf{L}}_R R')$ is represented by $K^\bullet \otimes_R R'$. Then of course for every $a \in I$ we see that $a \otimes 1 : K^{-1} \otimes_R R' \to K^{-1} \otimes_R R'$ is equal to $(h_a \otimes 1) \circ (d_K^{-1} \otimes 1)$. Since the collection of maps $K^{-1} \otimes_R R' \to K^{-1} \otimes_R R'$ which factor through $\mathrm{d}_K^{-1} \otimes 1$ forms an $R'$-module we conclude. $\square$